UNIT 1

Q.1 Describe basic architecture of Database Management System in detail. **(RGPV NOV 2023)**

Q.2 Explain the Functions of database Administrator. **(RGPV NOV 2023)**

Q.3 What is Database Management System (DBMS)? What are the major components of this system? Explain each component with neat sketch. **(RGPV NOV 2022)**

Q.4 What are the different Data Modules of DBMS? Explain in detail. **(RGPV NOV 2022)**

UNIT 2

Q.1 What is ER-model? Explain the basic symbols used of entities, attributes and relationships. **(RGPV NOV 2023)**

Q.2 What is an Attribute? Explain various types of attribute with examples. **(RGPV NOV 2023)**

Q.3 Explain integrity constraint over relations. **(RGPV NOV 2023)**

Q.4 Explain Fundamental operations in relational algebra with examples. **(RGPV NOV 2023)**

Q.5 Explain strong and weak entities. How weak entity can be converted to a strong entity. **(RGPV NOV 2022)**

Q.6 Explain Super key, primary key and candidate key with examples. **(RGPV NOV 2022)**

Q.7 What are integrity constraints? Explain various types of integrity constraints with suitable example. **(RGPV NOV 2022)**

Q.8 Discuss about Relational algebra operations like Select, Project, Join with examples. **(RGPV NOV 2022)**

UNIT 3

Q.1 How to alter, destroy and views table? Explain with suitable queries. **(RGPV NOV 2023)**

Q.2 What are the steps to be followed to convert a relation in 3NF to BCNF? **(RGPV NOV 2023)**

Q.3 Explain Optimization methods. **(RGPV NOV 2023)**

i) Heuristic based

ii) Cost Estimation based

Q.4 Define join dependency. How is it different to that of Multivalued dependency and Functional dependency? Justify. **(RGPV NOV 2022)**

Q.5 Define Normalization. Explain second normal form with the help of an example, **(RGPV NOV 2022)**

Q.6 Explain why 4NF is more desirable than BCNF with the help of an example. **(RGPV NOV 2022)**

UNIT 4

Q.1 How to test Serializability of Schedule? Explain with example. **(RGPV NOV 2023)**

Q.2 Explain log-based recovery Protocol. **(RGPV NOV 2023)**

Q.3 Explain Multiple Granularity Concurrency control scheme. **(RGPV NOV 2023)**

Q.4 What do you mean by transaction processing? Explain ACID properties of transaction in brief. **(RGPV NOV 2022)**

Q.5 How concurrency is performed? Explain the protocol that is used to maintain the concurrency concept. **(RGPV NOV 2022)**

Q.6 What is time stamping? Explain a mechanism of concurrency control that uses time stamping with the help of an examples. **(RGPV NOV 2022)**

UNIT 5

Q.1 What is view? Give brief note on view. **(RGPV NOV 2023)**

Q.2 State and explain various File Organization Methods. **(RGPV NOV 2023)**

Q.3 Write short notes on any two: **(RGPV NOV 2023)**

a) Assertions

b) Web and Mobile database

c) Hierarchical queries

d) Validation based protocols

Q.4 Explain the advantage of stored procedures with an example. **(RGPV NOV 2022)**

Q.5 What is Trigger? Explain in detail with an example. **(RGPV NOV 2022)**

Q.6 Write short notes on any two of the following: **(RGPV NOV 2022)**

a) Nested and Parameterized Cursors

b) Single level and multilevel indices

c) Domain Relational Calculus

— Best of Luck for Exam —